

Camila Paz Guajardo Vásquez

🌐 Camila Guajardo Vásquez | ✉️ cguajardo@dim.uchile.cl

PhD student

Department of Mathematical Engineering, Universidad de Chile.

I am interested in geometric topology, geometric analysis and hyperbolic geometry.

In my Master's thesis, I reimplemented a computational routine to obtain bounds on the systolic length and kissing number of compact Riemann surfaces, starting from an abstract group action.

Education

- PhD in Engineering Sciences, Mathematical Modeling 2026 – present
Institution: Faculty of Physical and Mathematical Sciences, Universidad de Chile
- Master of Science in Mathematics 2024 – 2026
Thesis: *On the action of the automorphism group of a compact Riemann surface on its set of systoles*
Supervisor: PhD. Anita María Rojas Rodríguez
Institution: Faculty of Sciences, Universidad de Chile
- Bachelor's Degree in Mathematics 2020 – 2023
Undergraduate thesis: *Mapping class groups of surfaces*
Supervisor: PhD. Mauricio Bustamante Londoño
Institution: Faculty of Mathematics, Pontificia Universidad Católica de Chile

Grants and Awards

- ANID National Doctorate Scholarship (Beca de Doctorado Nacional) 2026
- ANID National Master's Scholarship (Beca de Magíster Nacional) 2025
- Academic Excellence Award, Class of 2023 (Bachelor's in Mathematics) awarded in 2024

Research Experience

- Thesis Student, Fondecyt Project No. 12340034 2024
Topic: Group actions on compact Riemann surfaces and their effect on the set of closed geodesics.
Supervisor: Prof. Anita Rojas
- Undergraduate thesis 2023
Topic: *Mapping Class Group of surfaces*
Supervisor: Prof. Mauricio Bustamante Londoño
- Undergraduate Research Projects (Vicerrectoría de Investigación, PUC) 2022–2023
 - Riemann-Hurwitz formula and applications, with Prof. Natalia García Fritz
 - More theory and applications in topology, with Prof. Mauricio Bustamante Londoño
 - Geometry of plane curves, with Prof. Giancarlo Urzúa

Selected talks and Academic Activities

- Organizer, Student Lectures Series in Riemannian Geometry, PUC 2025 (Semester 2)
Topics: Smooth manifolds, tangent spaces, Riemannian metrics, model spaces, connections, geodesics, and curvature.
Faculty Advisor: Prof. Pedro Gaspar
- Student's seminar, Universidad de Talca 2025
Topic: Systolic bounds obtained from hyperbolic polygons

- Student's seminar, Universidad de Chile 2024–2025
Talks included *Concrete surfaces using SageMath* and *An Introduction to Morse theory*
- Number theory Seminar, PUC 2022–2023
Talks on additive fibers and elliptic surfaces over $\mathbb{P}^1$, and elliptic curves and their rational points

Selected Coursework

- POST664–Analysis II (Prof. Nikola Kamburov) 6.4/7.0
Contents: Banach and Hilbert spaces, Baire and Hahn–Banach theorems, Lax–Milgram theorem, spectral theory, Sobolev spaces, distributions
- POST5900–Introduction to Riemann surfaces and algebraic curves (Prof. Anita Rojas) 7.0/7.0
Contents: Riemann surfaces, algebraic curves in $\mathbb{P}^2(\mathbb{C})$, meromorphic functions, holomorphic maps, Riemann–Hurwitz formula
- CSDMATE157–Complex tori (Prof. Robert Auffarth) 6.1/7.0
Contents: Torus homomorphisms, Néron–Severi group, dual tori, extensions, subtori complements
- CS07MM0019–Geometry and topology of manifolds (Prof. Mauricio Bustamante) 6.7/7.0
Contents: Smooth manifolds, immersions and submersions, Sard's theorem, transversality
- MAT2850–Algebraic topology (Prof. Siddharth Mathur) 6.3/7.0
Contents: Covering spaces, fundamental groups, singular and cellular homology
- MAT2305–Differential Geometry, con (Prof. Mauricio Bustamante) 6.4/7.0
Contents: Curves in $\mathbb{R}^3$, fundamental forms, curvature, geodesics, Gauss–Bonnet's theorem

Teaching experience (Selected)

- Instructor. Department of Mathematics, Universidad Andrés Bello
 - FMMP001–Mathematical thinking 2026
- Teaching Assistant. Faculty of Mathematics, Pontificia Universidad Católica de Chile
 - MPG3403–Geometry and topology of manifolds, Prof. Pedro Gaspar 2025
 - MAT2704/MAT270I–Complex variables I, Prof. Martin Chuaqui 2024
 - MAT1389–Geometry I, Profs. María José Moreno and María Fernanda Espinal 2023
 - MAT2218–Abstract Algebra II, Prof. Ricardo Menares 2023
 - MAT1204–Introduction to Algebra, Prof. Amal Taarabt 2022
- Teaching Assistant. Department of Mathematics, Universidad de Chile
 - CVER004–Algebra and Geometry II, Prof. Anita Rojas 2025
 - MCLM110/MC110–Algebra in Geometry I, Prof. Nelda Jaque Tamblay 2024

Outreach and service

- Mathematics Festival Monitor, Puerto de Ideas Antofagasta 2023
- Coach, Chilean team for International Mathematical Olympiads 2021–present
- Problem grader, Chilean National Mathematical Olympiad 2021–present
- Member, Academic Committee CMat (Campeonato de Matemáticas) 2020–2022
- Problem Coordinator, 1st Pan–American Girls' Mathematical Olympiad 2021

Mathematical Olympiads' Participation

• Ibero-American MO, Bronze Medal	Mexico, 2019
• International MO, Honorable Mention	UK, 2019
• European Girls' MO, Bronze Medal	Ukraine, 2019
• International MO	Romania, 2018
• South Cone MO, Honorable Mention	Ecuador, 2017
• International MO	Brazil, 2017